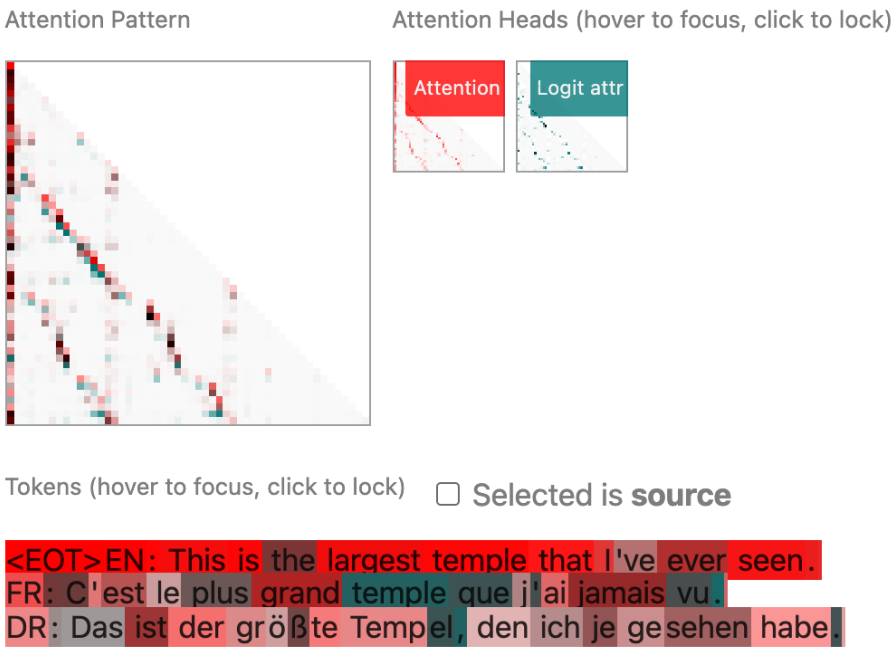


Note that the overall attention pattern (in red, top left) is more or less an “off-diagonal”, but it meanders a little bit away from a sharp diagonal. The meandering is because different languages have different word order and token lengths. As this head attends sequentially to past tokens that will semantically come next, the attended token position in the earlier sentences jumps around.

The logit attribution patterns for this head are not perfectly sharp; that is, even in cases where the attention head is attending to the matching word in an earlier language, it does not always *directly* increase the logit of the corresponding prediction. We would guess that this is because this head’s output needs to be further processed by later layers. However, taken overall, the direct logit attributions show clear evidence of contributing on net to the correct translation.

Behavior 3: Pattern matching

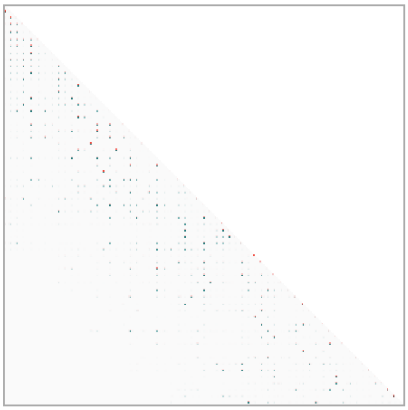
In this final example, we show an attention head (found at layer 26 of our 40-layer model) which does more complex pattern matching. One might even think of it as learning a simple function in context! (Again, this head *also* scores highly on our **measurements** of “basic” induction behavior when shown repeated random sequences, so it is an induction head by that definition.)

To explore this behavior, we've generated some synthetic text which follows a simple pattern. Each line follows one of four templates, followed by a label for which template it is drawn from. The template is random selected, as are the words which fill in the template:

- (month) (animal): 0
- (month) (fruit): 1
- (color) (animal): 2
- (color) (fruit): 3

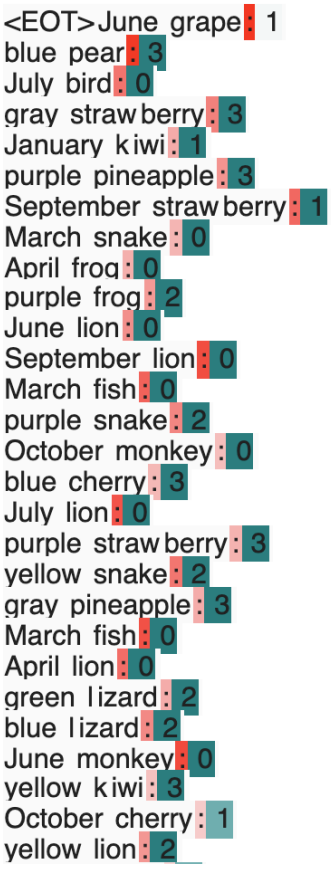
Below, we show how the attention head behaves on this synthetic example. To make the diagram easier to read, we've masked the attention pattern to only show the ":" tokens are the destination, and the logit attribution to only show where the output is the integer tokens.

Attention Pattern



Attention Heads (hover to focus, click to lock)

Tokens (hover to focus, click to lock) ☐ Selected is **source**



This head attends back to a previous instance of the correct category more often than not. It often knows to skip over lines where one of the words is identical but the pattern is wrong (such as "January bird" primarily attending to "April fish" and not "grey bird"). This head isn't perfect at this, but empirically it allocates about 65% of its attention from the colons to the correct positions, when tested on a range of similar problems.

What's going on with more abstract heads that are also induction heads?

We emphasize again that the attention heads that we described above simultaneously implement *both* the abstract behaviors that we described, and these *very same* attention heads (as in, the exact same head in the same layer) *also* satisfy the formal definition of induction head (literal copying of random sequences using prefix matching). The comparison is not a metaphor or a blurring of the definition: induction heads which are *defined* by their ability to copy literal sequences *turn out to also* sometimes match more abstract patterns. This is what the table at the beginning of the section shows empirically.

But this still leaves the question: *why* do the same heads that inductively copy random text also exhibit these other behaviors? One hint is that these behaviors can be seen as "spiritually similar" to copying. Recall that where an induction head is defined as implementing a rule like `[A][B] … [A] → [B]`, our empirically observed heads also do something like `[A*][B*] … [A] → [B]` where `A*` and `B*` are similar to `A` and `B` in some higher-level representation. There are several ways these similar behaviors could be connected. For example, note that the first behavior is a special case of the second, so perhaps induction heads are implementing a more general algorithm that reverts to the special case of copying when given a repeated sequence[22]. Another possibility is that induction heads implement literal copying when they take a path through the residual stream that includes only them, but implement more abstract behaviors when they process the outputs of earlier layers that create more abstract representations (such as representations where the same word in English and French are embedded in the same place).

In Argument 5 we'll strengthen this argument by giving a mechanistic account of how induction heads (when doing simple copying with prefix-matching) attend back to the token that comes next in the pattern, and observe that the actual mechanism they use could naturally generalize to more abstract pattern matching. Our point in this section is just that it's actually quite natural for these more abstract induction heads to also exhibit the basic copying behaviors underlying our definition.